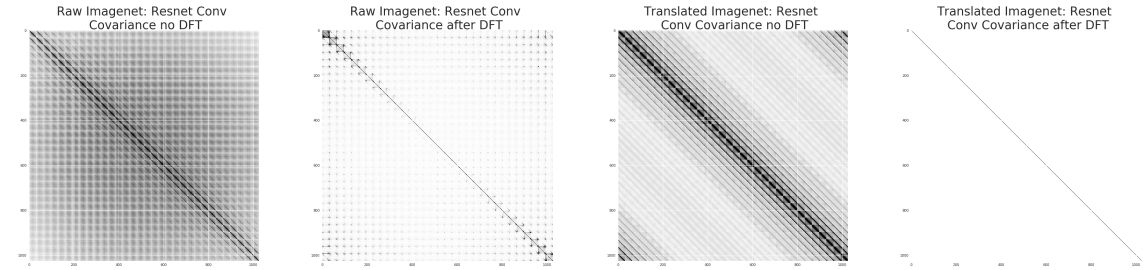

(a) (b) (c) (d)

Figure App.2: This figure visualizes the covariance matrix of one of the channels of a resnet trained on Imagenet. Black correspond to large values and white to small values. *(a)* we compute the covariance without a translation invariant dataset and without first preprocessing the images by DFT. We see that the covariance matrix is dense. *(b)* We compute the covariance after applying DFT, but without augmenting the dataset with translations. Even without enforcing translation invariance, we see that the covariance in the DFT basis is approximately diagonal. *(c)* Same as (a), but the dataset is augmented to be fully translation invariant. The covariance in the pixel basis is still dense. *(d)* Same as (c), but with dataset augmented to be translation invariant. The covariance matrix is exactly diagonal for a translation invariant dataset in a DFT basis.

When computing the covariance matrix however, we vectorize c by stacking the columns under each other, and call the result $vec(c)$:

$$vec(c) = \begin{bmatrix} \boldsymbol{z}_{0,0} \\ \boldsymbol{z}_{1,0} \\ \vdots \\ \boldsymbol{z}_{n-1,0} \\ \boldsymbol{z}_{0,1} \\ \vdots \\ \boldsymbol{z}_{n-1,n-1} \end{bmatrix} := \begin{bmatrix} \boldsymbol{z}_{0} \\ \boldsymbol{z}_{1} \\ \vdots \\ \boldsymbol{z}_{n-1} \\ \boldsymbol{z}_{n} \\ \vdots \\ \boldsymbol{z}_{n^2-1} \end{bmatrix}$$

One useful identity when switching between these two notations (see e.g. [8]) is

$$vec(AcB) = (B^T \otimes A)vec(c)$$

where A, B are matrices and $\otimes$ is the Kronecker product. A useful observation arising from this is:

Lemma 3. *The CCA vectors of $DFT(c_i), DFT(c_j)$ are the same (up to a rotation by F) as the CCA of c_i, c_j.*

Proof: From Section B we know that unitary transforms only rotate the CCA directions. But while DFT pre and postmultiplies by F, F^T – unitary matrices, we cannot directly apply this as the result is for unitary transforms on $vec(c_i)$. But, using the identity above, we see that $vec(DFT(c_i)) = vec(Fc_iF^T) = (F \otimes F)vec(c_i)$, which is unitary as F is unitary. Applying the same identity to c_j, we can thus conclude that the DFT preserves CCA (up to rotations).

As Theorem 1 preprocesses the neurons with DFT, it is important to note that by the Lemma above, we do not change the CCA vectors (except by a rotation).

C.1 Proof of Lemma 1

Proof. Translation invariance is preserved We show inductively that any translation invariant input to a convolutional channel results in a translation invariant output: Suppose the input to channel c, (n by n) is translation invariant. It is sufficient to show that for inputs X_i, X_j and $0 \leq a, b, \leq n-1$, $c(X_i) + (a, b) \mod n = c(X_j)$. But an (a, b) shift in neuron coordinates in c corresponds to a (height stride $\cdot\, a$, width stride $\cdot\, b$) shift in the input. And as X is translation invariant, there is some $X_j = X_i +$ (height stride $\cdot\, a$, width stride $\cdot\, b$).

$cov(c)$ is circulant:

Let X be (by proof above) a translation invariant input to a channel c in some convolution or pooling layer. The empirical covariance, $cov(c)$ is the n^2 by n^2 matrix computed by (assuming c is centered)

$$\frac{1}{|X|}\sum_{X_i \in X} vec(c(X_i)) \cdot vec(c(X_i))^T$$

So, $cov(c)_{ij} = \frac{1}{|X|}\boldsymbol{z}_i^T \boldsymbol{z}_j = \frac{1}{|X|}\sum_{X_l \in X} \boldsymbol{z}_i^T(X_l)\boldsymbol{z}_j(X_l)$, i.e. the inner products of the neurons i and j.

The indexes i and j refer to the neurons in their vectorized order in $vec(c)$. But in the matrix ordering of neurons in c, i and j correspond to some (a_1, b_1) and (a_2, b_2). If we applied a translation (a, b), to both, we would get new neuron coordinates $(a_1 + a, b_1 + b), (a_2 + a, b_2 + b)$ (all coordinates $\bmod n$) which would correspond to $i + an + b \mod n^2$ and $j + an + b \mod n^2$, by our stacking of columns and reindexing.

Let $\tau_{a,b}$ be the translation in inputs corresponding to an (a, b) translation in c, i.e. $\tau_{a,b} =$ (height stride$\cdot a$, width stride$\cdot b$). Then clearly $\boldsymbol{z}_{(a_1,b_1)}(X_i) = \boldsymbol{z}_{(a_1+a,b_1+b)}(\tau_{(a,b)}(X_i)$, and similarly for $\boldsymbol{z}_{(a_2,b_2)}$

It follows that $\frac{1}{|X|}\boldsymbol{z}_{(a_1,b_1)}^T\boldsymbol{z}_{(a_2,b_2)} = \frac{1}{|X|}\boldsymbol{z}_{(a_1+b,b_1+b)}^T\boldsymbol{z}_{(a_2+a,b_2+b)}$, or, with $vec(c)$ indexing

$$\frac{1}{|X|}\boldsymbol{z}_i^T\boldsymbol{z}_j = \frac{1}{|X|}\boldsymbol{z}_{(i+an+b \mod n^2)}^T\boldsymbol{z}_{(j+an+b \mod n^2)}$$

This gives us the circulant structure of $cov(c)$.

$cov(c)$ is block circulant: Let $\boldsymbol{z}^{(i)}$ be the ith column of c, and $\boldsymbol{z}^{(j)}$ the jth. In $vec(c)$, these correspond to $\boldsymbol{z}_{(i-1)n}, \ldots \boldsymbol{z}_{in-1}$ and $\boldsymbol{z}_{(j-1)n}, \ldots \boldsymbol{z}_{jn-1}$, and the n by n submatrix at those row and column indexes of $cov(vec(c))$ corresponds to the covariance of column i, j. But then we see that the covariance of columns $i+k, j+k$, corresponding to the covariance of neurons $\boldsymbol{z}_{(i-1)n+k\cdot n}, \ldots \boldsymbol{z}_{in-1+k\cdot n}$, and $\boldsymbol{z}_{(j-1)n+k\cdot n}, \ldots \boldsymbol{z}_{jn-1+k\cdot n}$, which corresponds to the 2-d shift $(1, 0)$, applied to every neuron. So by an identical argument to above, we see that for all $0 \leq k \leq n-1$

$$cov(\boldsymbol{z}^{(i)}, \boldsymbol{z}^{(j)}) = cov(\boldsymbol{z}^{(i+k)}, \boldsymbol{z}^{(j+k)})$$

In particular, $cov(vec(c))$ is block circulant. □

An example $cov(vec(c))$ with c being 3 by 3 look like below:

$$\begin{bmatrix} A_0 & A_1 & A_2 \\ A_2 & A_0 & A_1 \\ A_1 & A_2 & A_0 \end{bmatrix}$$

where each A_i is itself a circulant matrix.

C.2 Proof of Lemma 2

Proof. This is a standard result, following from expressing a circulant matrix A in terms of its diagonal form , i.e. $A = V\Sigma V^T$ with the columns of V being its eigenvectors. Noting that $V = F$, the DFT matrix, and that vectors of powers of $\omega_k = \exp(\frac{2\pi ik}{n})$, $\omega_j = \exp(\frac{2\pi ik}{n})$ are orthogonal gives the result. □

C.3 Proof of Theorem 2

Proof. Starting with (a), we need to show that $cov(vec(DFT(c_i)), vec(DFT(c_i))$ is diagonal. But by the identity above, this becomes:

$$cov(vec(DFT(c_i)), vec(DFT(c_i)) = (F \otimes F)vec(c_i)vec(c_i)^T(F \otimes F)^*$$

By Lemma 1, we see that

$$cov(vec(c_i)) = vec(c_i)vec(c_i)^T = \begin{bmatrix} A_0 & A_1 & \dots & A_{n-1} \\ A_{n-1} & A_0 & \dots & A_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ A_1 & A_2 & \dots & A_0 \end{bmatrix}$$

with each A_i circulant.

And so $cov(vec(DFT(c_i)), vec(DFT(c_i))$ becomes

$$\begin{bmatrix} f_{00}F & f_{01}F & \dots & f_{0,n-1}F \\ f_{10}F & f_{11}F & \dots & f_{1,n-1}F \\ \vdots & \vdots & \ddots & \vdots \\ f_{n-1,0}F & f_{n-1,1}F & \dots & f_{n-1,n-1}F \end{bmatrix} \begin{bmatrix} A_0 & A_1 & \dots & A_{n-1} \\ A_{n-1} & A_0 & \dots & A_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ A_1 & A_2 & \dots & A_0 \end{bmatrix} \begin{bmatrix} f_{00}^*F^* & f_{10}^*F^* & \dots & f_{n-1,0}^*F^* \\ f_{01}^*F^* & f_{11}^*F^* & \dots & f_{n-1,1}^*F^* \\ \vdots & \vdots & \ddots & \vdots \\ f_{0,n-1}^*F^* & f_{1,n-1}^*F^* & \dots & f_{n-1,n-1}^*F^* \end{bmatrix}$$

From this, we see that the sjth entry has the form

$$\sum_{l=0}^{n-1} \left(\sum_{k=0}^{n-1} f_{sk} F A_{l-k} \right) f_{lj}^* F^* = \sum_{k,l} f_{sk} f_{lj}^* F A_{l-k} F^*$$

Letting $[FA_rF^*]$ denote the coefficient of the term FA_rF^*, we see that (addition being $\mod n$)

$$[FA_rF^*] = \sum_{k=0}^{n-1} f_{sk} f_{(k+r)j}^* = \sum_k e^{\frac{2\pi isk}{n}} \cdot e^{\frac{-2\pi ij(k+r)}{n}} = e^{\frac{-2\pi ijr}{n}} \sum_{k=0}^{n-1} e^{\frac{2\pi ik(s-j)}{n}} = e^{\frac{-2\pi ijr}{n}} \cdot \delta_{sj}$$

with the last step following by the fact that the sum of powers of non trivial roots of unity are 0.

In particular, we see that only the diagonal entries (of the n by n matrix of matrices) are non zero. The diagonal elements are linear combinations of terms of form FA_rF^*, and by Lemma 2 these are diagonal. So the covariance of the DFT is diagonal as desired.

Part (b) follows almost identically to part (a), but by first noting that exactly by the proof of Lemma 1, $cov(c_i, c_j)$ is also a circulant and block circulant matrix.

□

C.4 Proof of Theorem 1

Proof. This Theorem now follows easily from the previous. Suppose we have a layer l, with channels $c_1, ..., c_k$. And let $vec(DFT(c_i))$ have directions $\tilde{\boldsymbol{z}}_0^{(i)}, \cdots \tilde{\boldsymbol{z}}_{n^2-1}^{(i)}$. By the previous theorem, we know that the covariance of all of these neurons only has non-zero terms $cov(\tilde{\boldsymbol{z}}_k^{(i)}, \tilde{\boldsymbol{z}}_k^{(j)}$.

So arranging the full covariance matrix to have row and column indexes being $\tilde{\boldsymbol{z}}_0^{(1)}, \tilde{\boldsymbol{z}}_0^{(1)}, \dots \tilde{\boldsymbol{z}}_0^{(k)}, \tilde{\boldsymbol{z}}_1^{(1)} \dots \tilde{\boldsymbol{z}}_{n^2}^{(k)}$ the nonzero terms all live in the n^2 k by k blocks down the diagonal of the matrix, proving the theorem. □

C.5 Computational Gains

As the covariance matrix is block diagonal, our more efficient algorithm for computation is as follows: take the DFT of every channel ($n \log n$ due to FFT) and then compute covariances according to blocks: partition the k^n directions into the n^2 k by k matrices that are non-zero, and compute the covariance, inverses and square roots along these.

A rough computational budget for the covariance is therefore $kn \log n + n^2k^{2.5}$, while the naive computation would be of order $(kn^2)^{2.5}$, a polynomial difference. Furthermore, the DFT method also makes for easy parallelization as each of the n^2 blocks does not interact with any of the others.